$$u_t = \nabla^2 [u^3 - u - \gamma \nabla^2 u]$$ 4th order PDE bc 2 laplace.

$$= \nabla^2 \left[u^3 - u - \gamma \left(\frac{1}{r^2} [r^2 u_r]_r \right) \right]$$

outer lap: makes nonlocal.

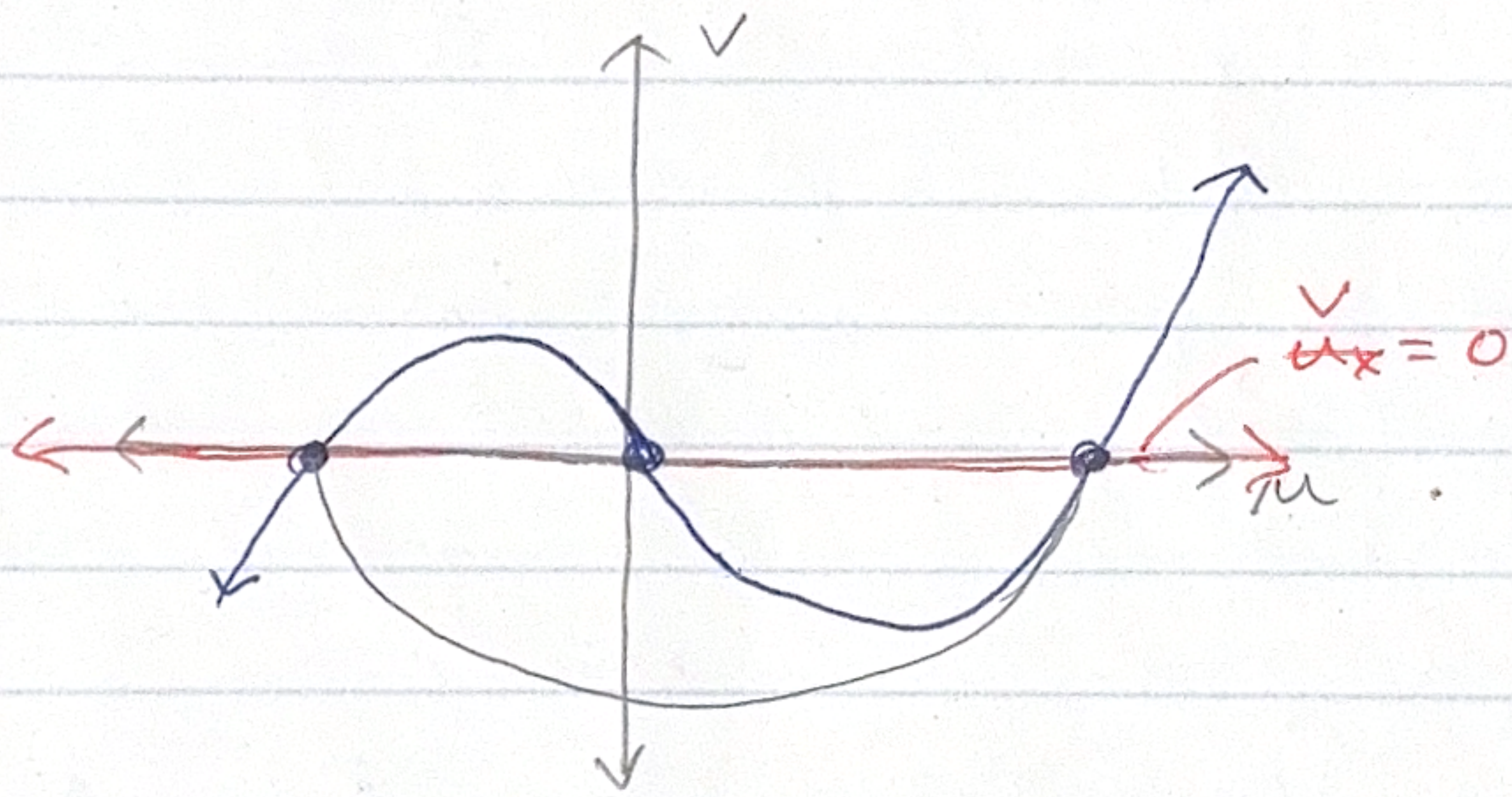
Cahn-Hilliard: $u_t = \nabla^2 [f'(u) - \gamma \nabla^2 u]$

$f'(u)$: $u^3 - u$ $\quad \gamma > 0$

$\nabla^2 u = \frac{1}{r^2} [r^2 u_r]_r$

~~pro~~ chain rule?

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \left[u^3 - u - \gamma \left(\frac{1}{r^2} \frac{d}{dr} (r^2 u_r) \right) \right] \right)$$



try: doing same steps but plug laplaci. into $w = f(u) - \gamma u_{xx}$ → see what happens.

$$\begin{cases} u_x = v \\ v_x = \frac{1}{\gamma} [\underbrace{f(u)}_{u^3 - u} - w] \end{cases}$$

Basic outline:

- we have steady-state equation that can be gererated from $w = 0$
- then can draw steady-state system → convert to single steady state equation → above graph to tan.